Física I –terceira Aula

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https://www.geogebra.org/m/z6hbxuns

Adição de vetores

$A = (1, 2, 3)$

$B = \text{Interseca}(\text{EjeY}, \text{EjeZ})$

$\rightarrow (0, 0, 0)$

$u = \text{Vector}(B, A)$

$\rightarrow \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$C = (20, 3, 7)$

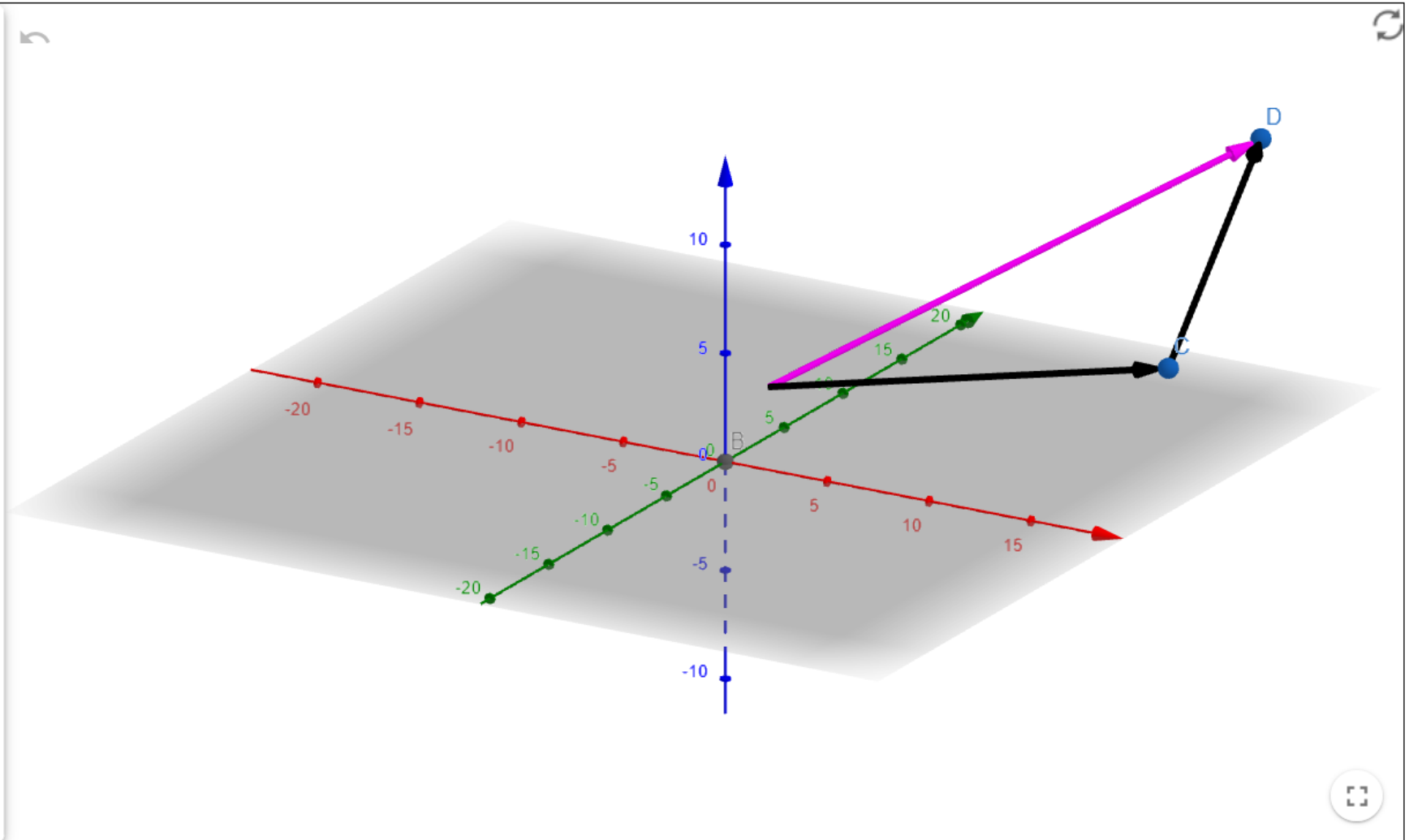
$D = (13, 23, 10)$

$w = \text{Vector}(C, D)$

$\rightarrow \begin{pmatrix} -7 \\ 20 \\ 3 \end{pmatrix}$

$v = \text{Vector}(A, C)$

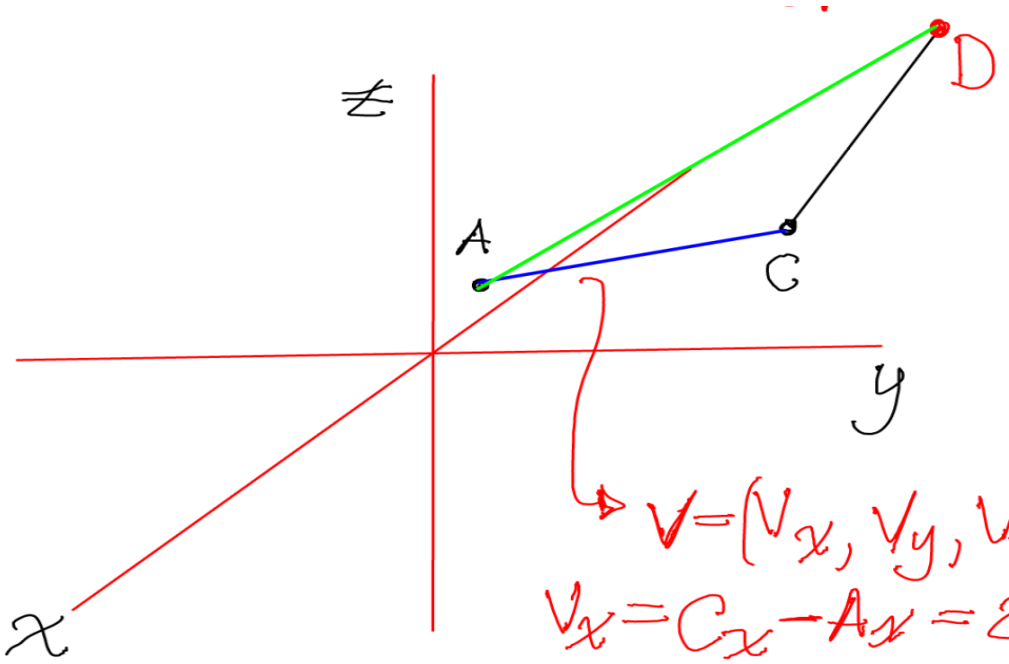
$\rightarrow \begin{pmatrix} 19 \\ 1 \\ 4 \end{pmatrix}$



$A = (1, 2, 3)$

$C = (20, 3, 7)$

$D = (13, 23, 10)$



$\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$

$\vec{V} = 19\hat{i} + \hat{j} + 4\hat{k}$

$V = (V_x, V_y, V_z)$

$V_x = C_x - A_x = 20 - 1 = 19$

$V_y = C_y - A_y = 1$

$V_z = C_z - A_z = 4$

$\vec{W} = (D_x - C_x)\hat{i} + (D_y - C_y)\hat{j} + (D_z - C_z)\hat{k}$

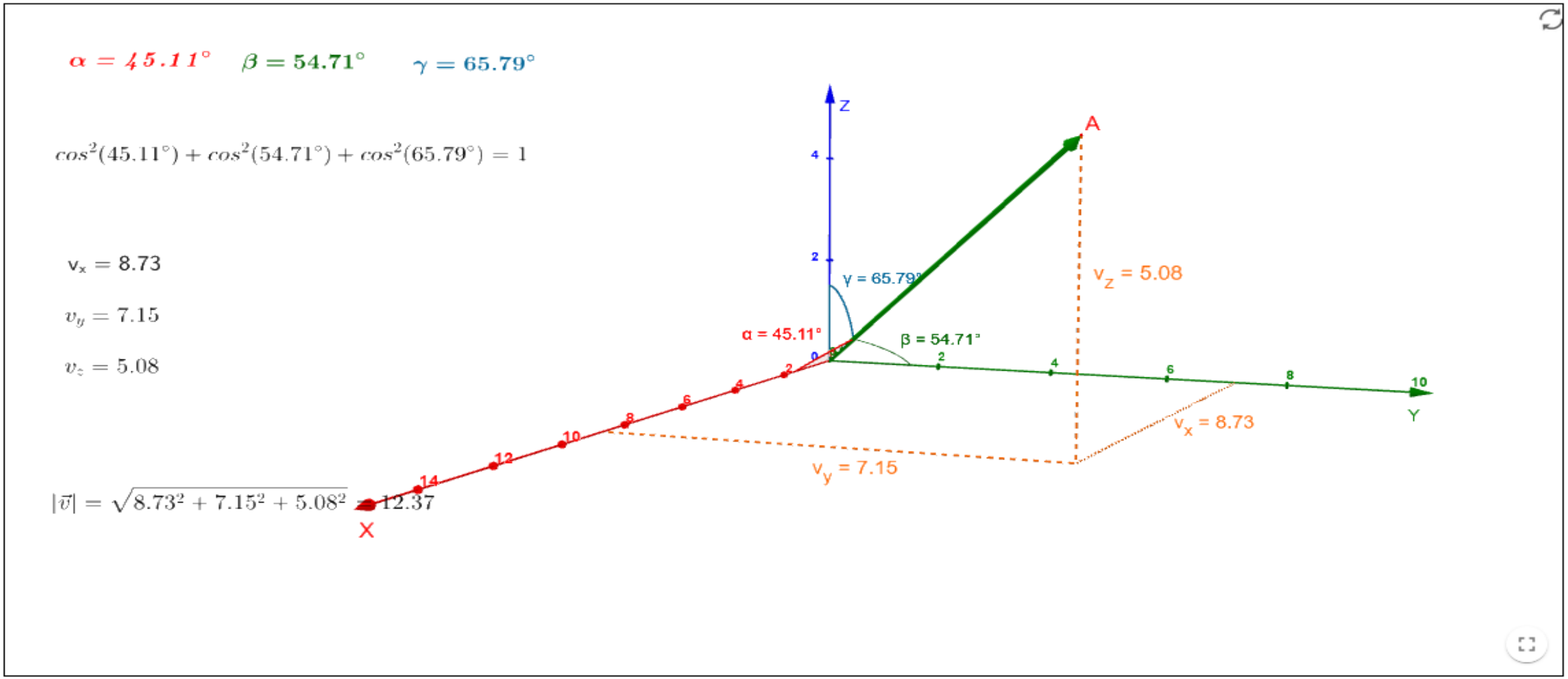
$\vec{W} = -7\hat{i} + 20\hat{j} + 3\hat{k}$

$\vec{S} = \vec{V} + \vec{W} = (V_x + W_x)\hat{i} + (V_y + W_y)\hat{j} + (V_z + W_z)\hat{k}$

$\boxed{\vec{S} = 12\hat{i} + 21\hat{j} + 7\hat{k}}$ ✓

Cossenos diretores de um vetor (3D)

https://www.geogebra.org/m/cYbhuwdp#material/PvPNNGb9



Exemplo

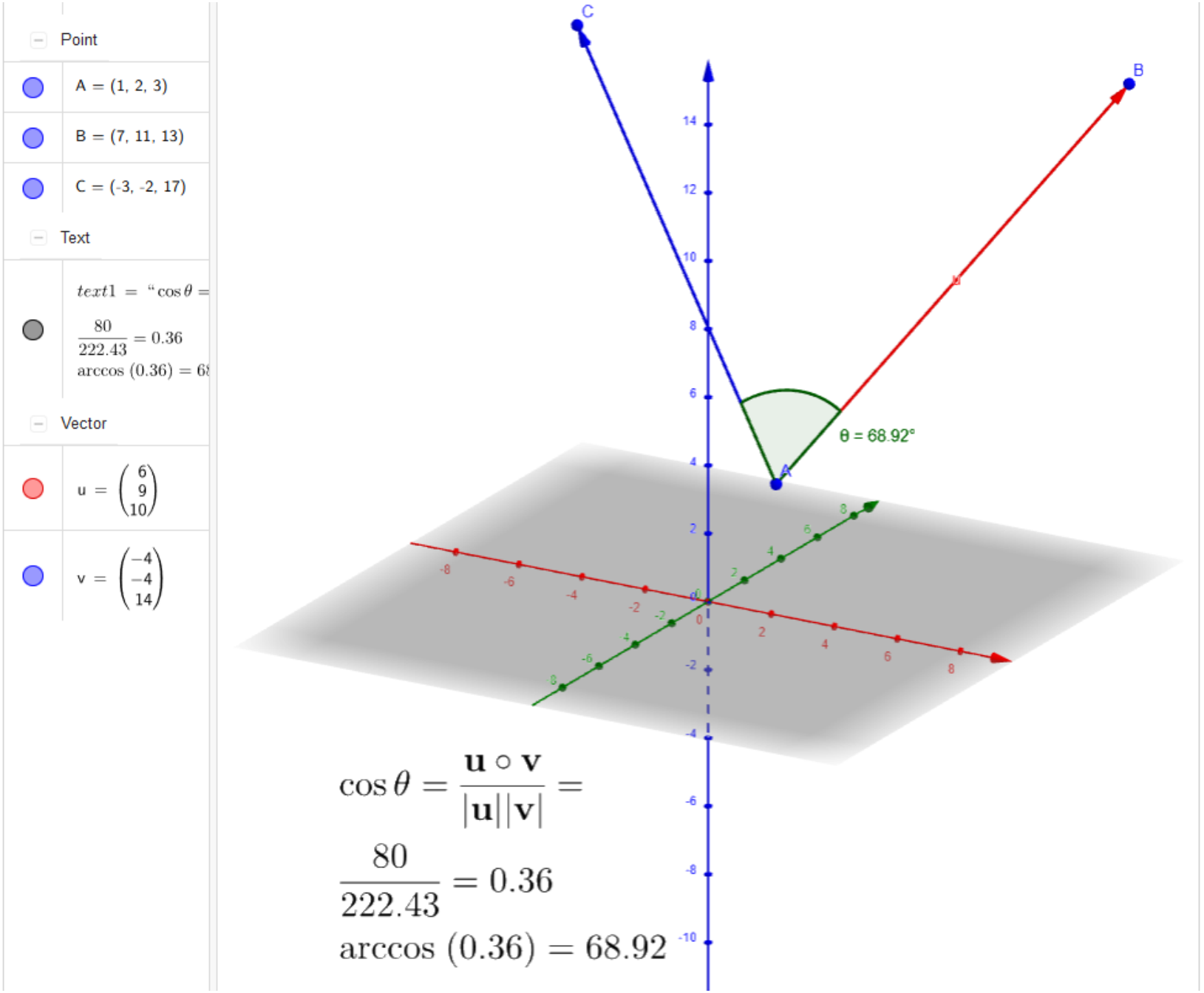
- Encontre os cossenos diretores do vetor que une os pontos $(3, 2, -4)$

$(1, -1, 2)$

$$\vec{A} = (3, 2, -4) - (1, -1, 2) = (2, 3, -6) \Longrightarrow \begin{cases} \cos\alpha = \dfrac{\hat{\imath} \cdot \vec{A}}{\left|\vec{A}\right|} = \dfrac{2}{7} \\ \cos\beta = \dfrac{\hat{\jmath} \cdot \vec{A}}{\left|\vec{A}\right|} = \dfrac{3}{7} \\ \cos\gamma = \dfrac{\hat{k} \cdot \vec{A}}{\left|\vec{A}\right|} = \dfrac{-6}{7} \end{cases}$$

Produto Escalar

https://www.geogebra.org/m/fjtrj8Cw

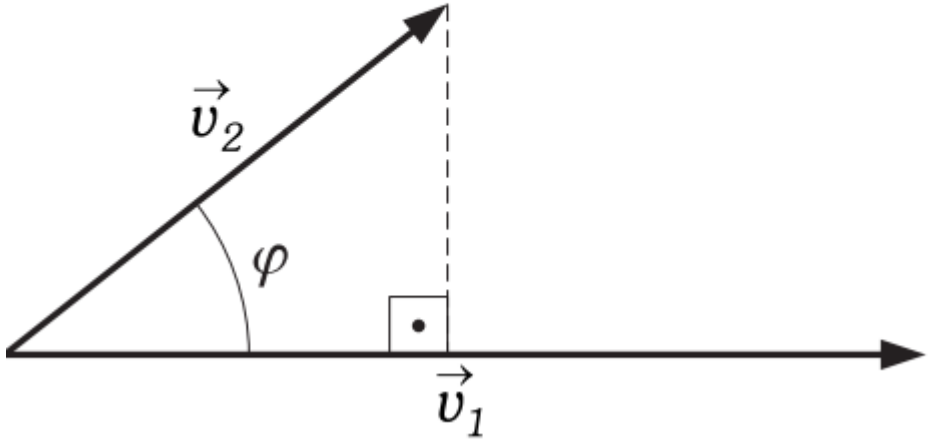


Produto Escalar de dois vetores

https://www.geogebra.org/m/fjtrj8Cw

O PRODUTO ESCALAR pode ser definido como um produto do módulo de um dos vetores pela projeção do outro sobre ele.

$$\vec{v}_1 \cdot \vec{v}_2 = v_1 v_2 \cos\varphi = v_1 \operatorname{proy}_{v_1} v_2$$



$$\boxed{\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = v_1 v_2 \cos\varphi}$$

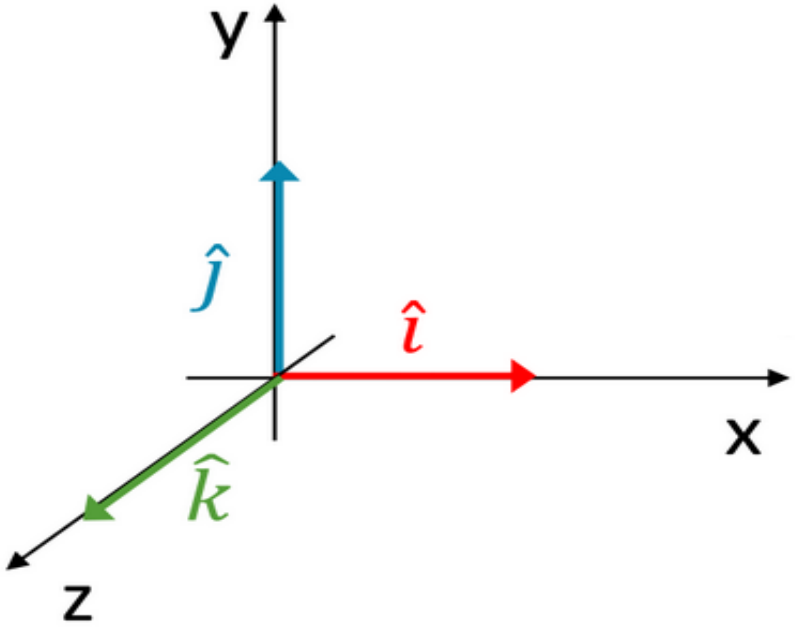
$$\boxed{\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = v_1 \operatorname{proy}_{v_1} v_2}$$

Se dois vetores são perpendiculares, seu produto escalar é zero

$$\left.\begin{array}{l} \boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = v_1 v_2 \cos\varphi \\ \varphi = \dfrac{\pi}{2} \quad \Rightarrow \quad \cos\varphi = 0 \end{array}\right| \quad \Rightarrow \quad \boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = 0$$

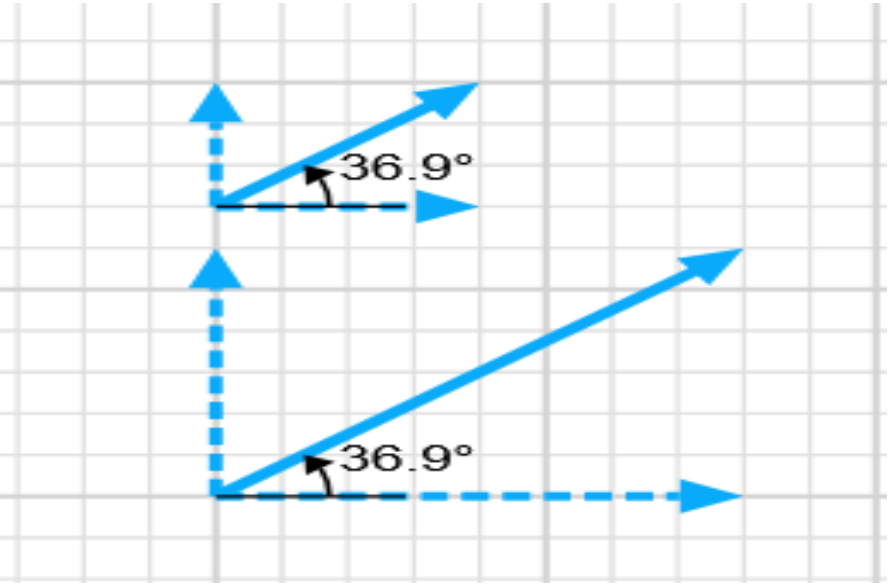
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = (x_1\boldsymbol{i} + y_1\boldsymbol{j} + z_1\boldsymbol{k}) \cdot (x_2\boldsymbol{i} + y_2\boldsymbol{j} + z_2\boldsymbol{k})$$

$$\boldsymbol{i}\cdot\boldsymbol{i} = \boldsymbol{j}\cdot\boldsymbol{j} = \boldsymbol{k}\cdot\boldsymbol{k} = 1, \;\; \boldsymbol{i}\cdot\boldsymbol{j} = \boldsymbol{j}\cdot\boldsymbol{k} = \boldsymbol{k}\cdot\boldsymbol{i} = 0,$$



Se dois vetores têm a mesma direção e sentido, seu produto escalar é igual ao produto de seus módulos

$$\varphi = 0 \qquad \cos\varphi = 1 \;\Rightarrow\; \boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = v_1 v_2$$



- Em particular, se relacionarmos **v** a si mesmo por meio do produto escalar, podemos escrever:

$$\boldsymbol{v} \cdot \boldsymbol{v} = v^2$$

Com base na propriedade distributiva vamos obter a expressão do produto escalar de dois vetores em função de seus componentes.

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = (x_1\boldsymbol{i} + y_1\boldsymbol{j} + z_1\boldsymbol{k}) \cdot (x_2\boldsymbol{i} + y_2\boldsymbol{j} + z_2\boldsymbol{k})$$

$$\boxed{\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = x_1x_2 + y_1y_2 + z_1z_2}$$

Ângulo de dois vetores: A propriedade anterior nos fornece um método simples para obter o ângulo formado por dois vetores, conhecendo seus componentes.

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = x_1x_2 + y_1y_2 + z_1z_2$$

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = v_1v_2 \cos\varphi$$

$$\Rightarrow \boxed{\cos\varphi = \frac{x_1x_2 + y_1y_2 + z_1z_2}{\sqrt{x_1^2 + y_1^2 + z_1^2}\,\sqrt{x_2^2 + y_2^2 + x_2^2}}} \Leftrightarrow \boxed{\cos\varphi = \frac{\boldsymbol{v}_1 \cdot \boldsymbol{v}_2}{v_1v_2}}$$

Exemplo:

- Encontre o vetor unitário paralelo ao plano OYZ e perpendicular vetor $\boldsymbol{v} = 2\boldsymbol{i} + \boldsymbol{j} - 3\boldsymbol{k}$.
- Solução:
- Vamos chamar **s** ao nosso vetor, como é unitário devemos ter que |**s**|=1. Já que o vetor **s** está conteudo no plano OYZ sua componente em x é zero.

$$\mathbf{s} = \hat{s} = 0\mathbf{i} + s_y\mathbf{j} + s_z\mathbf{k}$$

$$\mathbf{s} \perp \mathbf{v}$$

$$\mathbf{s} \cdot \mathbf{v} = 0$$

$$(0\mathbf{i} + s_y\mathbf{j} + s_z\mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} - 3\mathbf{k}) = 0 \quad \longrightarrow \quad s_y - 3s_z = 0$$

$$s_y = 3s_z$$

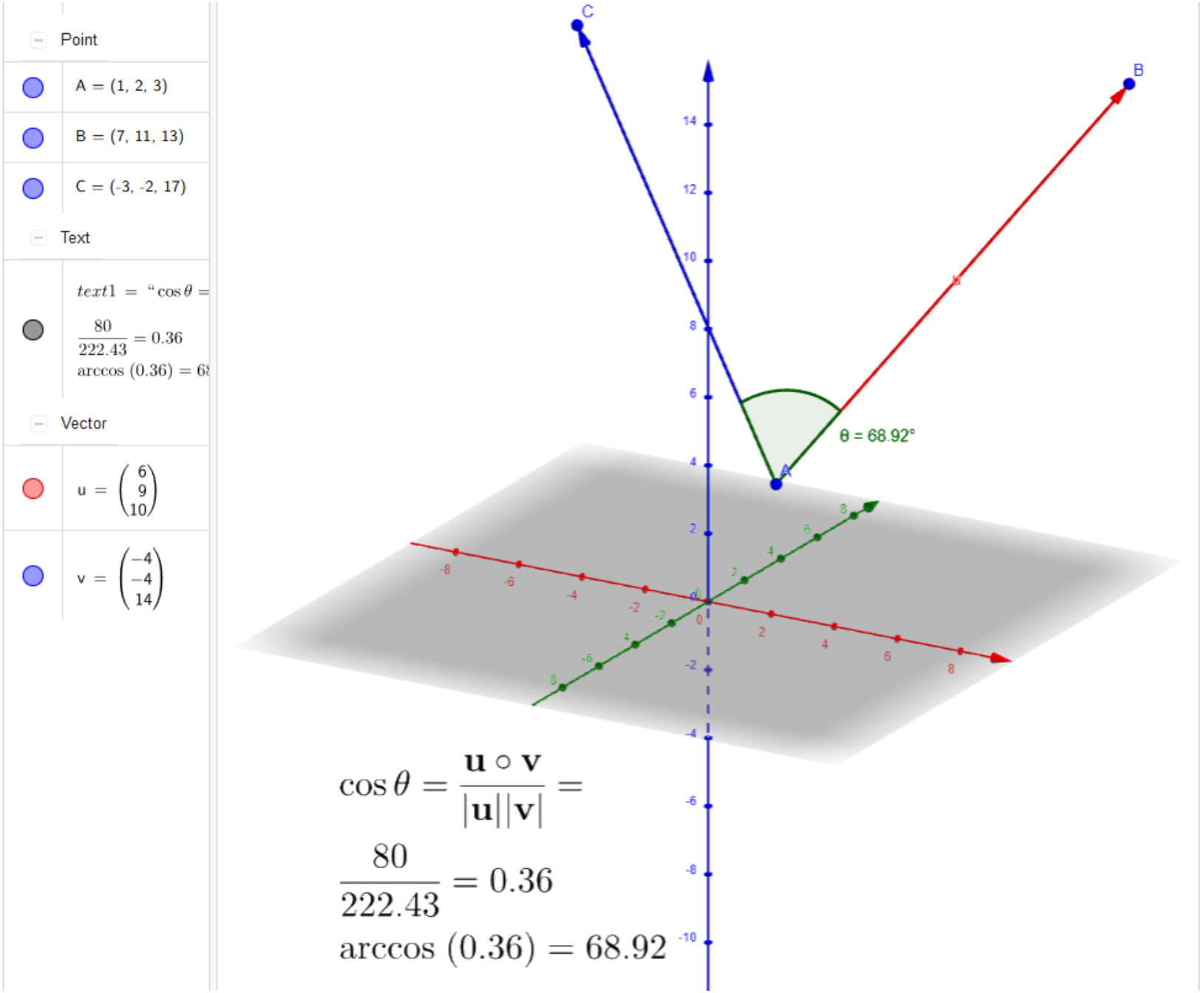
$$\mathbf{s} \cdot \mathbf{s} = (3s_z\mathbf{j} + s_z\mathbf{k}) \cdot (3s_z\mathbf{j} + s_z\mathbf{k}) = 1 \quad \longrightarrow \quad 9s_z^2 + s_z^2 = 1$$

$$s_z = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}$$

$$\mathbf{s} = \frac{\sqrt{10}}{10}(3\mathbf{j} + \mathbf{k})$$

Produto Escalar

https://www.geogebra.org/m/fjtrj8Cw

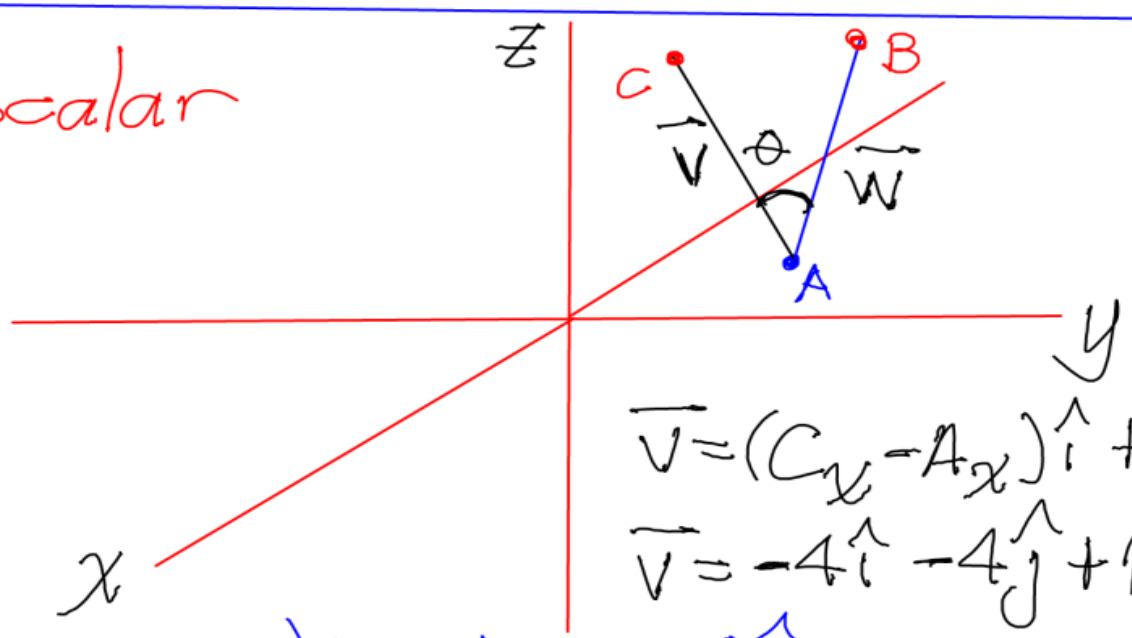


Produto Escalar

$A = (1, 2, 3)$

$C = (-3, -2, 17)$

$B = (7, 11, 13)$



$$\vec{V} = (C_x - A_x)\hat{i} + (C_y - A_y)\hat{j} + (C_z - A_z)\hat{k}$$

$$\vec{V} = -4\hat{i} - 4\hat{j} + 14\hat{k}$$

$$\vec{W} = (B_x - A_x)\hat{i} + (B_y - A_y)\hat{j} + (B_z - A_z)\hat{k}$$

$$\vec{W} = 6\hat{i} + 9\hat{j} + 10\hat{k}$$

$$\vec{V} \cdot \vec{W} = (-4\hat{i} - 4\hat{j} + 14\hat{k}) \cdot (6\hat{i} + 9\hat{j} + 10\hat{k}) = -24 - 36 + 140$$

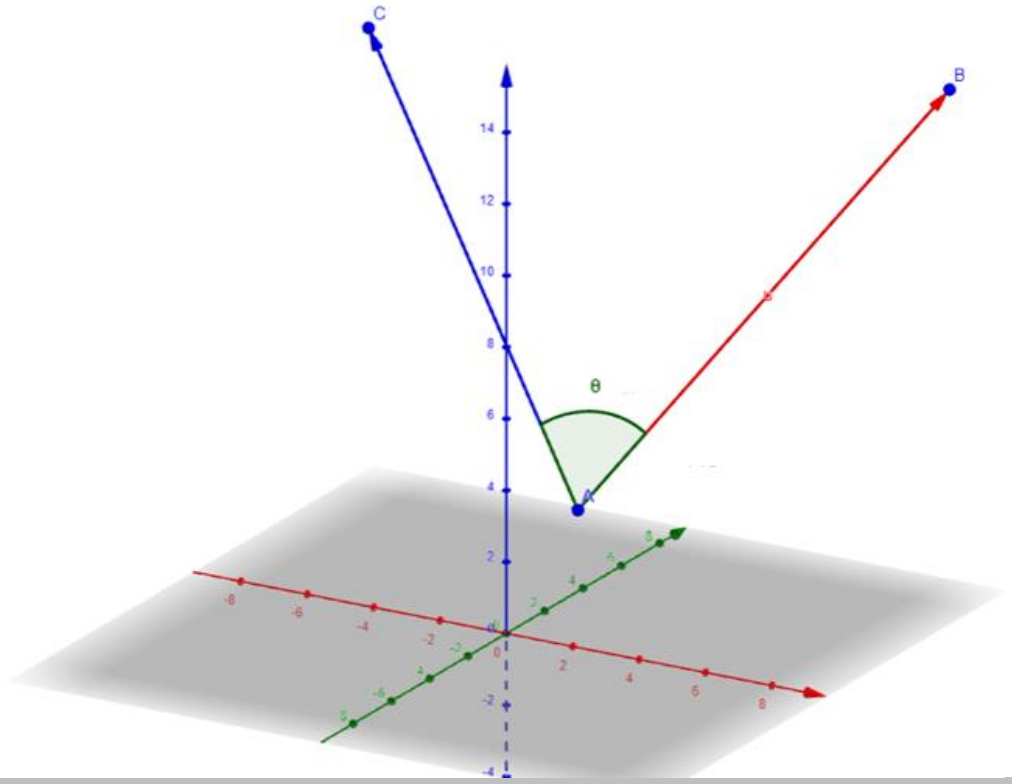
$$= -60 + 140$$

$$= 80$$

$$\vec{V} \cdot \vec{W} = |\vec{V}||\vec{W}| \cos\theta$$

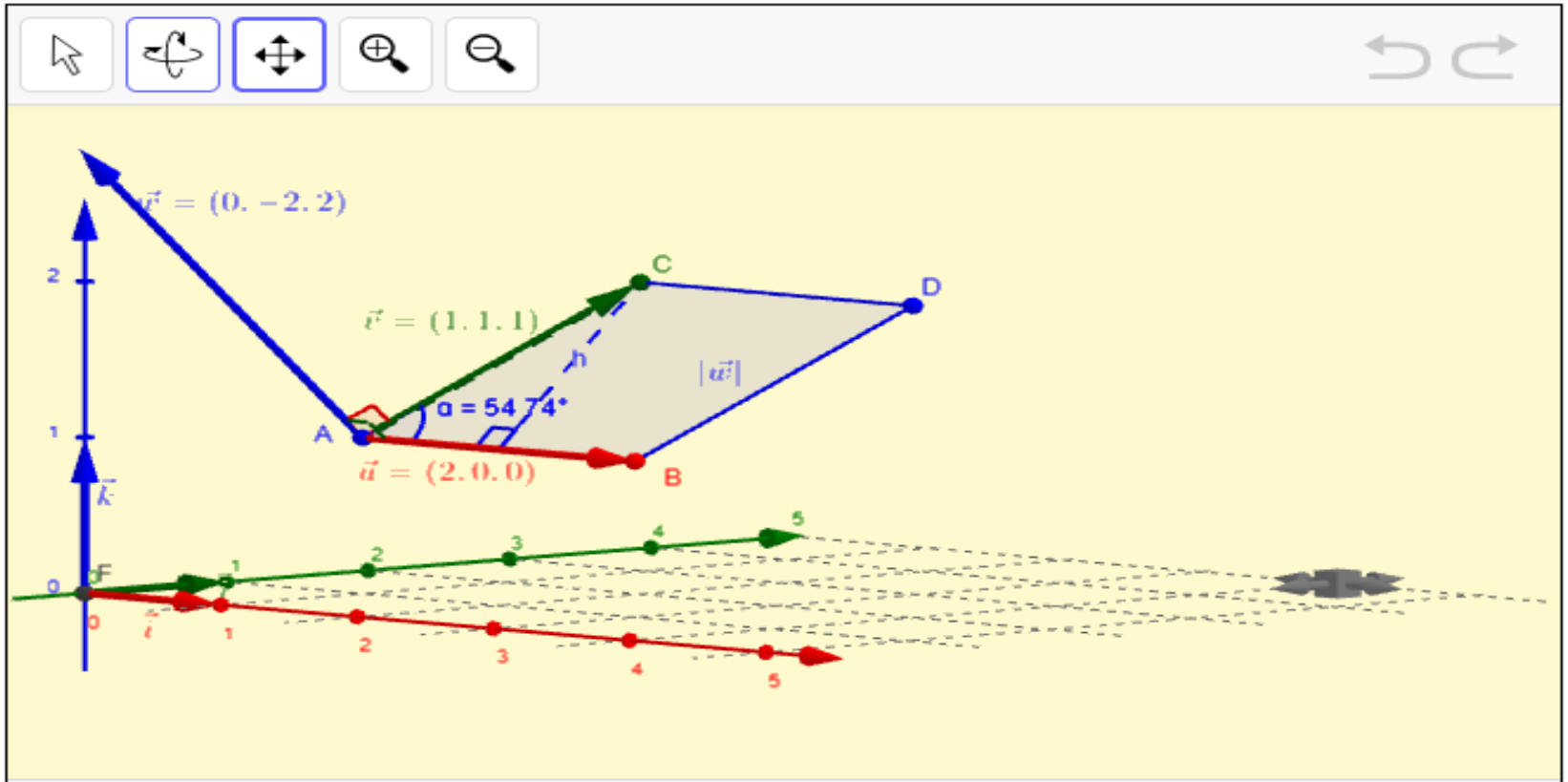
$$\cos\theta = \frac{\vec{v}\cdot\vec{w}}{|\vec{v}|\,|\vec{w}|} \; ; \quad |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2} = \sqrt{16 + 16 + (14)^2}$$

$$|\vec{w}| = \sqrt{w_x^2 + w_y^2 + w_z^2} = \sqrt{36 + 81 + 100}$$



Produto Vetorial

https://www.geogebra.org/m/TDsyaheC



A = (1, 1, 1)

B = (3, 1, 1)

C = (2, 2, 2)

$$\vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 0\,\vec{i} - 2\vec{j} + 2\vec{k}$$

Int. geom.

$$sen(\vec{u}, \vec{v}) = sen(54.74°) = 0.82$$

$$\vec{i} \times \vec{j} = -\vec{j} \times \vec{i} = \vec{k}$$
$$\vec{j} \times \vec{k} = -\vec{k} \times \vec{j} = \vec{i}$$
$$\vec{k} \times \vec{i} = -\vec{i} \times \vec{k} = \vec{j}$$

$$|\vec{u}| = \overrightarrow{AB} = \sqrt{u_1^2 + u_2^2 + u_3^2} = \sqrt{2^2 + 0^2 + 0^2} = \sqrt{4} = 2$$

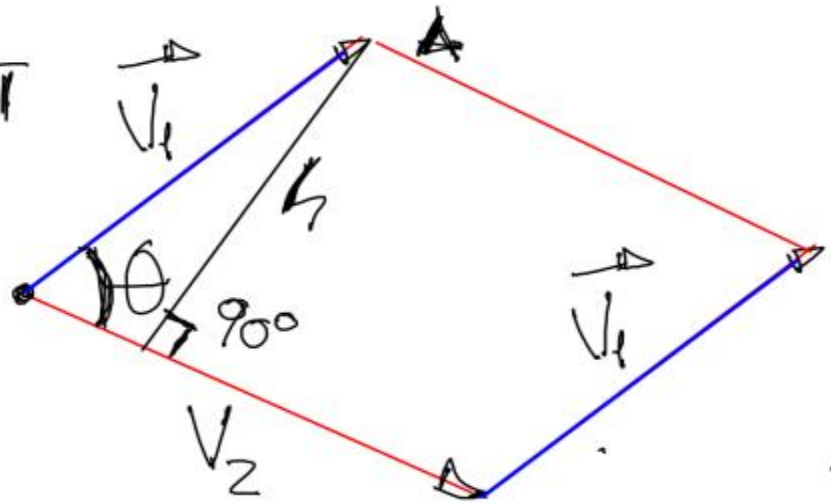
$$|\vec{v}| = \overrightarrow{AC} = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \cong 1.73$$

$$\vec{w} = \vec{u} \times \vec{v} : \begin{cases} |\vec{w}| = |\vec{u}| \cdot |\vec{v}| \cdot |sen(\vec{u}, \vec{v})| = 2 \cdot 1.73 \cdot |0.82| = 2.83 = |\vec{u}| \cdot h = Área(ABDC) \\ \vec{w} \perp \vec{u}, \quad \vec{w} \perp \vec{v} \\ \uparrow w : regla\ sacacorchos(\vec{u}, \vec{v}) \end{cases}$$

$$|\vec{w}| = \sqrt{w_1^2 + w_2^2 + w_3^2} = \sqrt{0^2 + (-2)^2 + 2^2} = \sqrt{8} \cong 2.83$$

$$\operatorname{sen}\theta = \frac{h}{|\vec{v}_1|}$$

$$|\vec{v}_1| \operatorname{sen}\theta = h$$



$$A = |\vec{v}_2|\, h$$

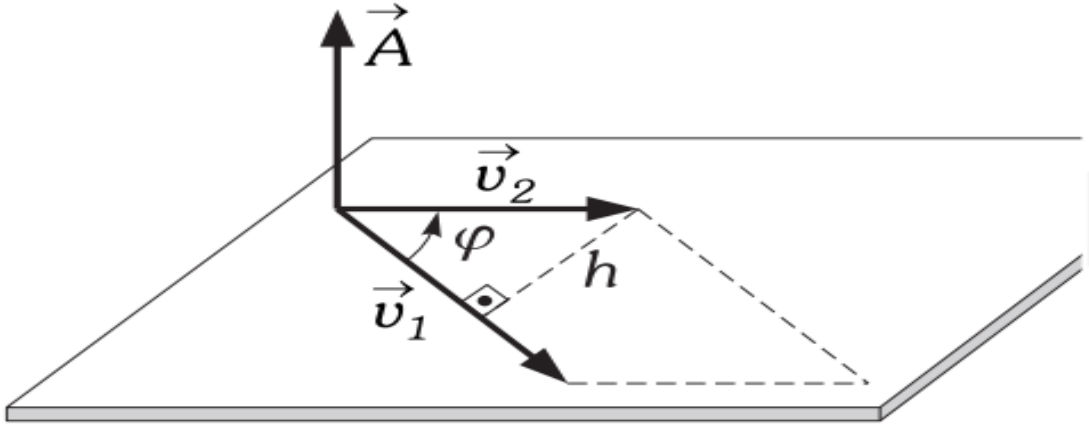
$$A = |\vec{v}_2|\, |\vec{v}_1| \operatorname{sen}\theta$$

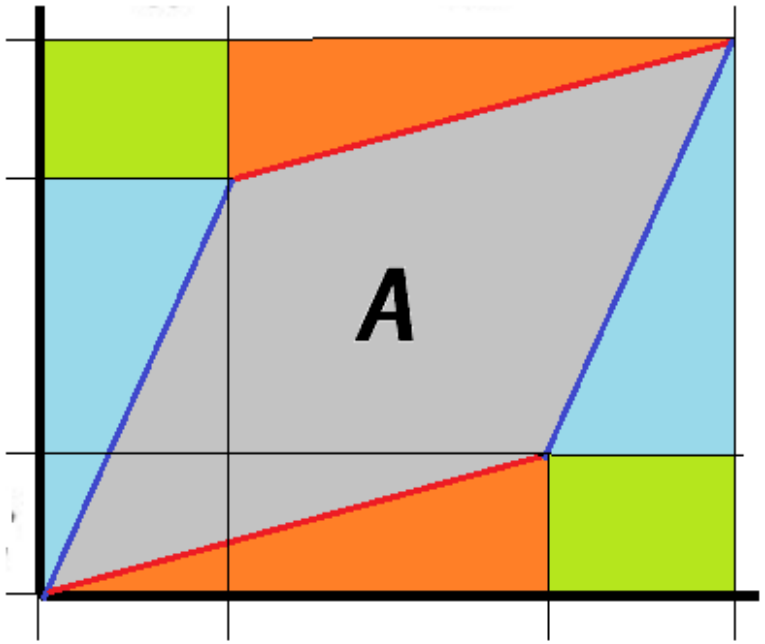
$$A = |\vec{v}_1 \times \vec{v}_2|$$

$$\vec{A} = \vec{V}_1 \times \vec{V}_2$$

$$\vec{A} = \vec{v}_1 \times \vec{v}_2$$

$$A = v_1 v_2 \operatorname{sen}\varphi$$





$$\vec{C} = \vec{a} \times \vec{b} \; ; \quad \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \begin{matrix} \hat{i} & \hat{j} \\ a_x & a_y \\ b_x & b_y \end{matrix} = \hat{i} a_y b_z + \hat{j} a_z b_x + \hat{k} a_x b_y - \hat{i} a_z b_y - \hat{j} a_x b_z - \hat{k} a_y b_x$$

$$\vec{C} = \hat{i}(a_y b_z - a_z b_y) + \hat{j}(a_z b_x - a_x b_z) + \hat{k}(a_x b_y - a_y b_x)$$

$$\vec{c} = \hat{i}(a_y b_z - a_z b_y) + \hat{j}(a_z b_x - a_x b_z) + \hat{k}(a_x b_y - a_y b_x)$$

Exemplo

$$\vec{a} = \overset{a_x}{4}\hat{i} + \overset{a_y}{3}\hat{j} + \overset{a_z}{2}\hat{k}$$

$$\vec{b} = \underset{b_x}{2}\hat{i} + \underset{b_y}{1}\hat{j} + \underset{b_z}{4}\hat{k}$$

$$\vec{c} = (12-2)\hat{i} + (4-16)\hat{j} + (4-6)\hat{k}$$

$$\vec{c} = 10\hat{i} - 12\hat{j} - 2\hat{k}$$

$$|\vec{c}| = |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta_{ab}$$

$$\sin\theta_{ab} = \frac{|\vec{c}|}{|\vec{a}||\vec{b}|}$$

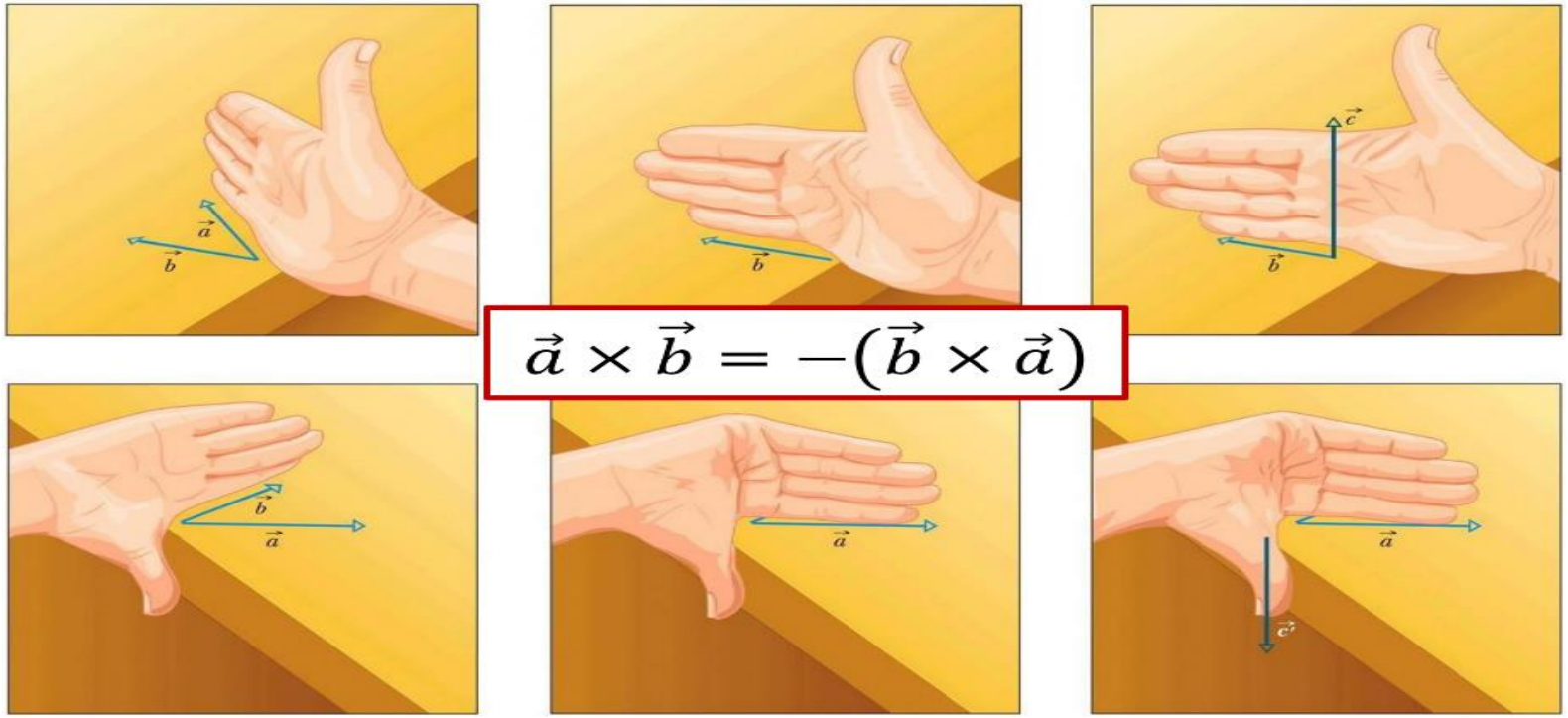
$$|\vec{a}| = \sqrt{16+9+4}$$

$$|\vec{b}| = \sqrt{4+1+16}$$

$$|\vec{c}| = \sqrt{100+144+4}$$

Propriedades do produto vetorial

- O produto vetorial possui propriedade anticomutativa (não possui propriedade comutativa), pois: $\boldsymbol{v}_1 \times \boldsymbol{v}_2 = -\boldsymbol{v}_2 \times \boldsymbol{v}_1$



$\hat{i} = 1, \quad \hat{j} = 2, \quad \hat{k} = 3$

$\hat{j} \times \hat{k} = \hat{i} \Rightarrow 231231$

$\hat{k} \times \hat{j} = -\hat{i} \Rightarrow 321321$

$\hat{i} \times \hat{j} = \hat{k} \Rightarrow 123123$

$\hat{j} \times \hat{i} = -\hat{k} \Rightarrow 213213$

$\hat{k} \times \hat{i} = \hat{j} \Rightarrow 312312$

$\hat{i} \times \hat{k} = -\hat{j} \Rightarrow 132132$

Exemplo:

- Dados dois vetores **a=** (1, -1, 2) e **b =**(-1, 3, 4), encontre um vetor unitário que seja perpendicular a ambos.

$$\begin{vmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \Rightarrow \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ -1 & 3 & 4 \end{vmatrix} \begin{matrix} i & j \\ 1 & -1 \\ -1 & 3 \end{matrix}$$

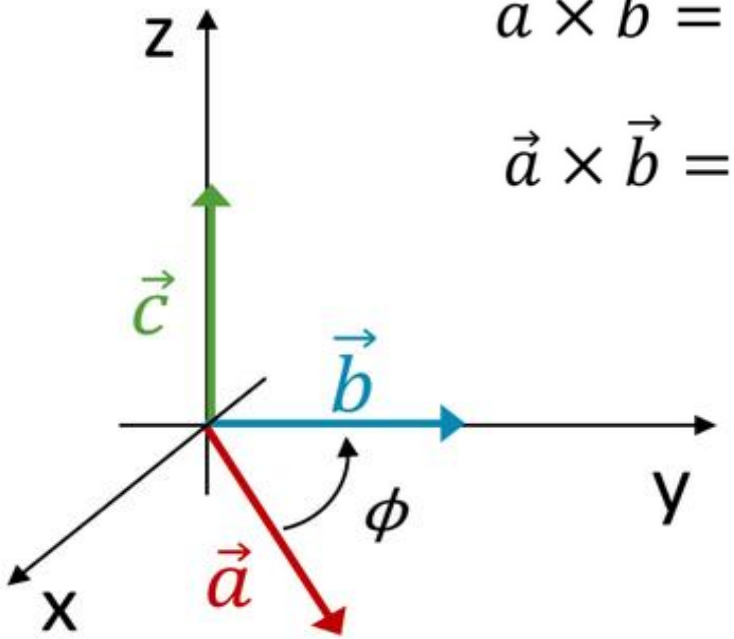
$$- \quad - \quad - \qquad + \quad + \quad +$$

$$\mathbf{c} = \mathbf{a} \times \mathbf{b} = -10\mathbf{i} - \mathbf{6j} + \mathbf{2k}$$

$$c^2 = \mathbf{c} \cdot \mathbf{c} = 100 + 36 + 4 = 140$$

$$\hat{c} = \frac{\mathbf{c}}{c} = \frac{1}{\sqrt{140}}(-10\mathbf{i} - \mathbf{6j} + \mathbf{2k})$$

Conceito de área como vetor



$$\vec{a} \times \vec{b} = \vec{c}$$

$$\vec{a} \times \vec{b} = \underbrace{(a_y b_z - b_y a_z)}_{c_x}\hat{\imath} + \underbrace{(a_z b_x - b_z a_x)}_{c_y}\hat{\jmath} + \underbrace{(a_x b_y - b_x a_y)}_{c_z}\hat{k}$$

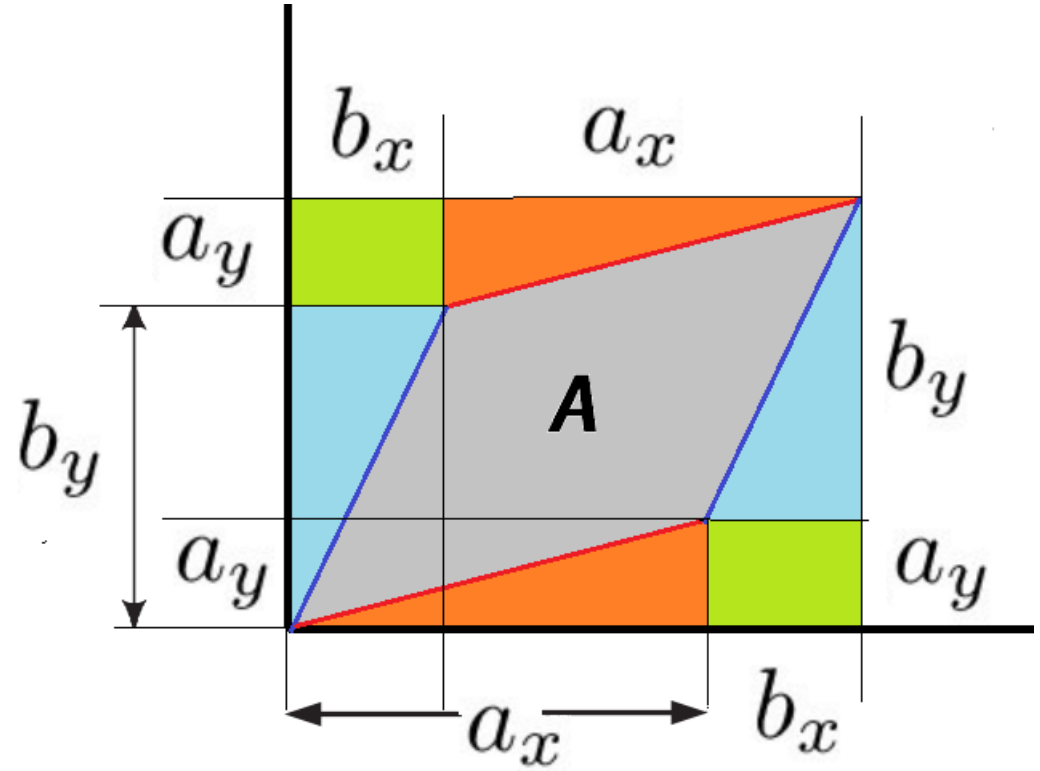
*Caso particular

$$c_x = c_y = 0$$

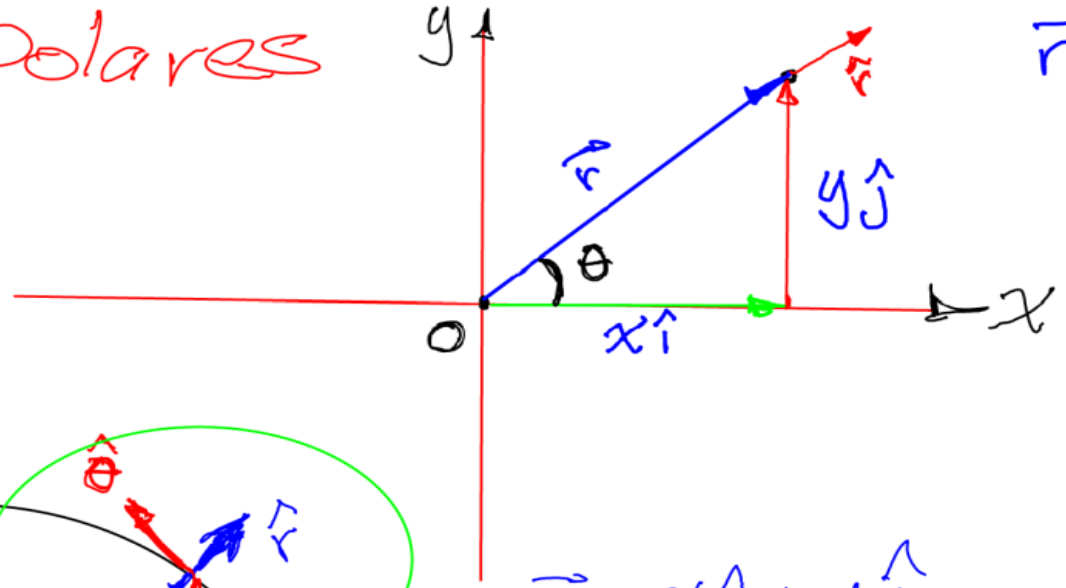
$$(a_x + b_x)(a_y + b_y) = a_x a_y + b_x b_y + 2 b_x a_y + A$$

$$a_x a_y + a_x b_y + b_x a_y + b_x b_y = a_x a_y + b_x b_y + 2 b_x a_y + A$$

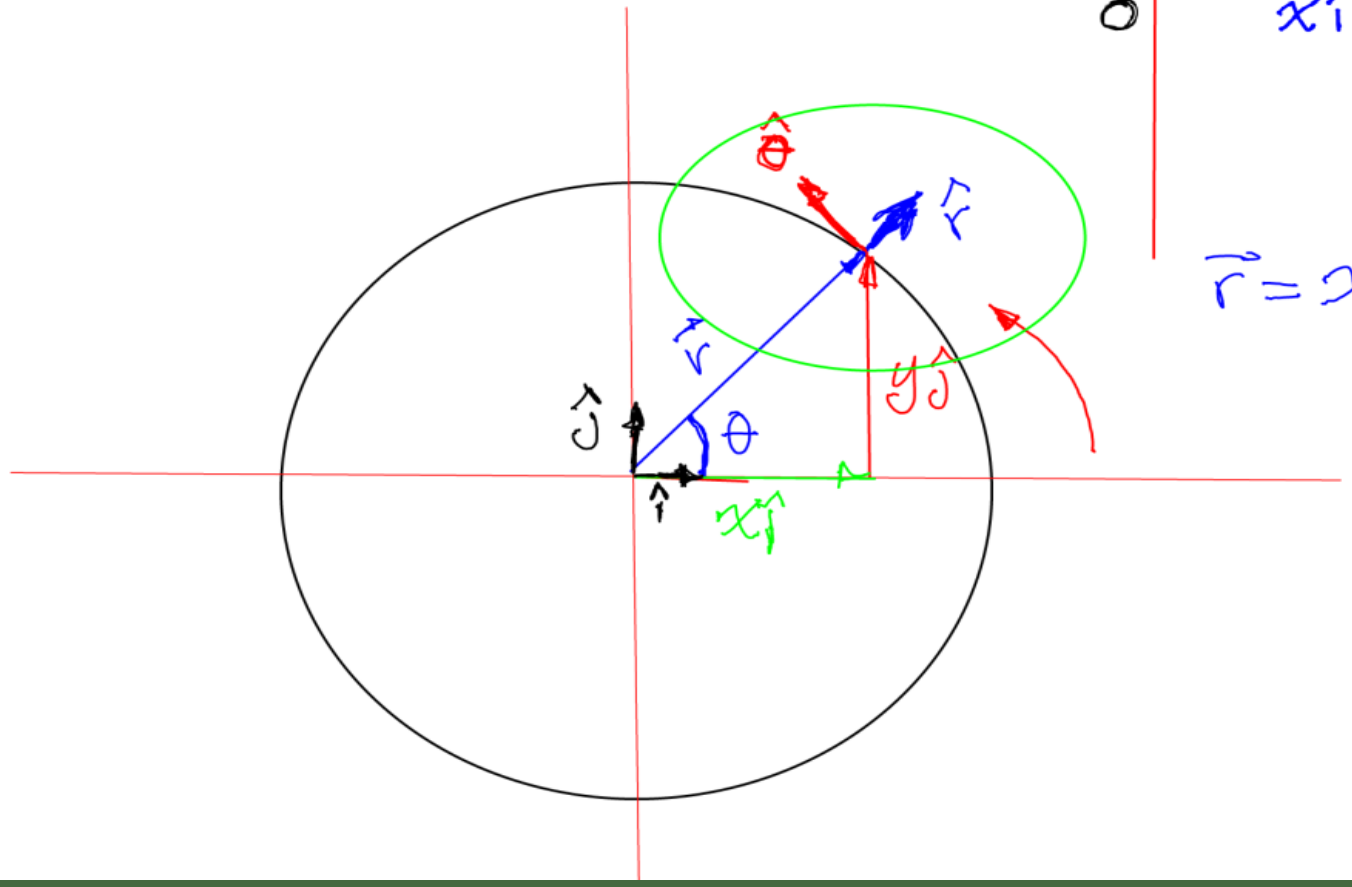
$$a_x b_y - b_x a_y = A$$



Coordenadas Polares



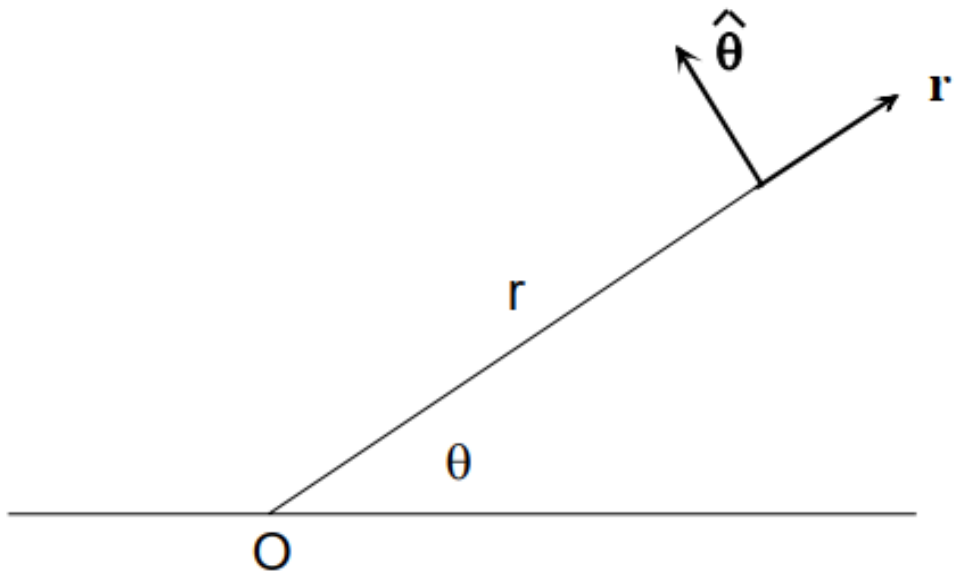
$$\vec{r} = x\hat{i} + y\hat{j}$$



$$\vec{r} = x\hat{i} + y\hat{j} = r\hat{r}$$

Resumo de Coordenadas polares

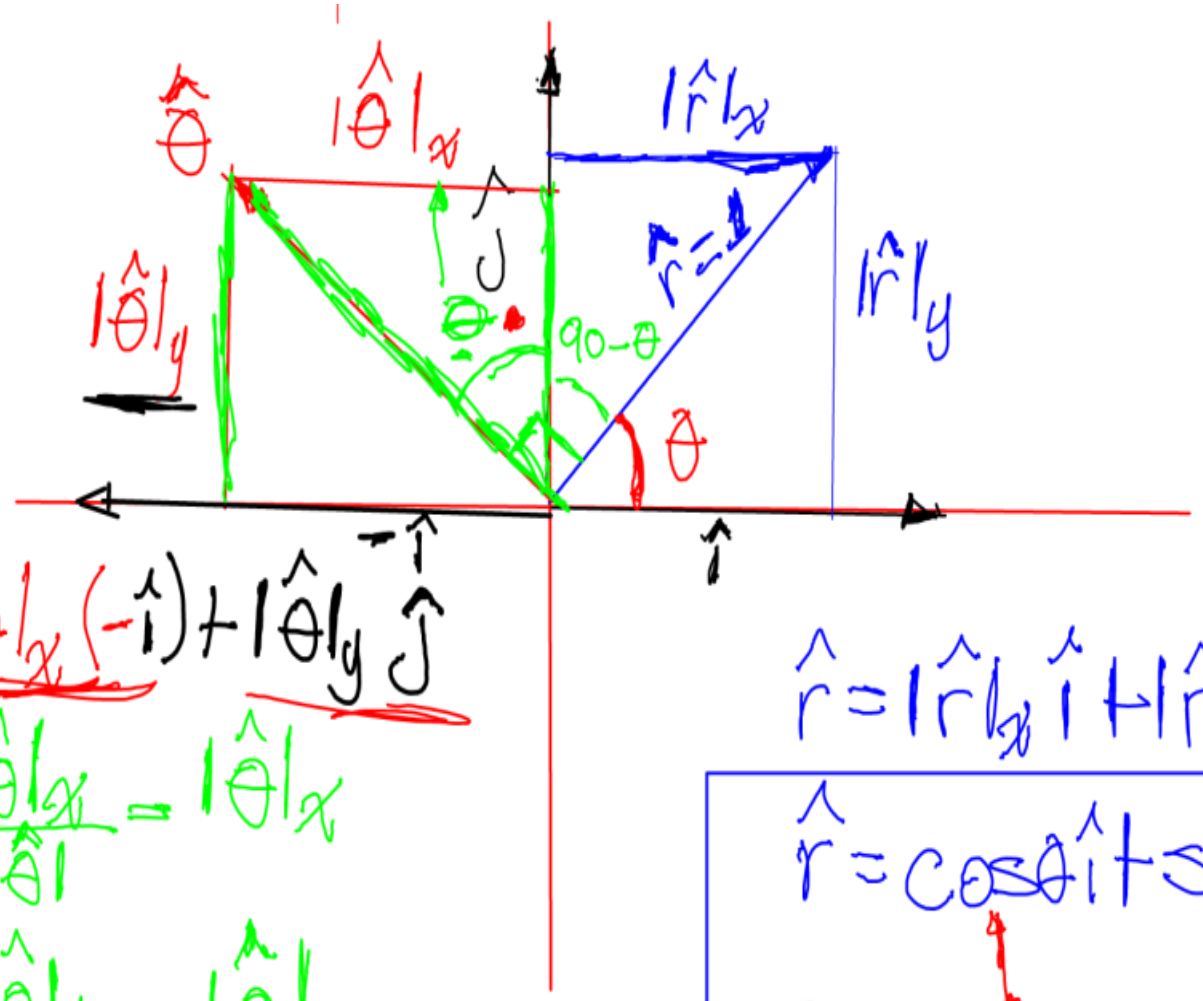
$$\begin{aligned}\hat{r} &= \hat{\imath}\cos\theta + \hat{\jmath}\sin\theta,\\ \hat{\theta} &= -\hat{\imath}\sin\theta + \hat{\jmath}\cos\theta.\end{aligned}$$



$$\begin{aligned}\frac{d\hat{r}}{dt} &= \dot{\theta}\left(-\hat{\imath}\sin\theta + \hat{\jmath}\cos\theta\right) = \dot{\theta}\hat{\theta},\\ \frac{d\hat{\theta}}{dt} &= -\dot{\theta}\left(\hat{\imath}\cos\theta + \hat{\jmath}\sin\theta\right) = -\dot{\theta}\hat{r}.\end{aligned}$$

velocidade angular

$$\frac{d\hat{r}}{dt} = \dot{\theta}\hat{k} \times \hat{r}$$



$$\sin\theta = \frac{|\hat{r}|_y}{|\hat{r}|} = |\hat{r}|_y$$

$$\cos\theta = \frac{|\hat{r}|_x}{1} = |\hat{r}|_x$$

$$\hat{\theta} = |\hat{\theta}|_x(-\hat{i}) + |\hat{\theta}|_y\,\hat{j}$$

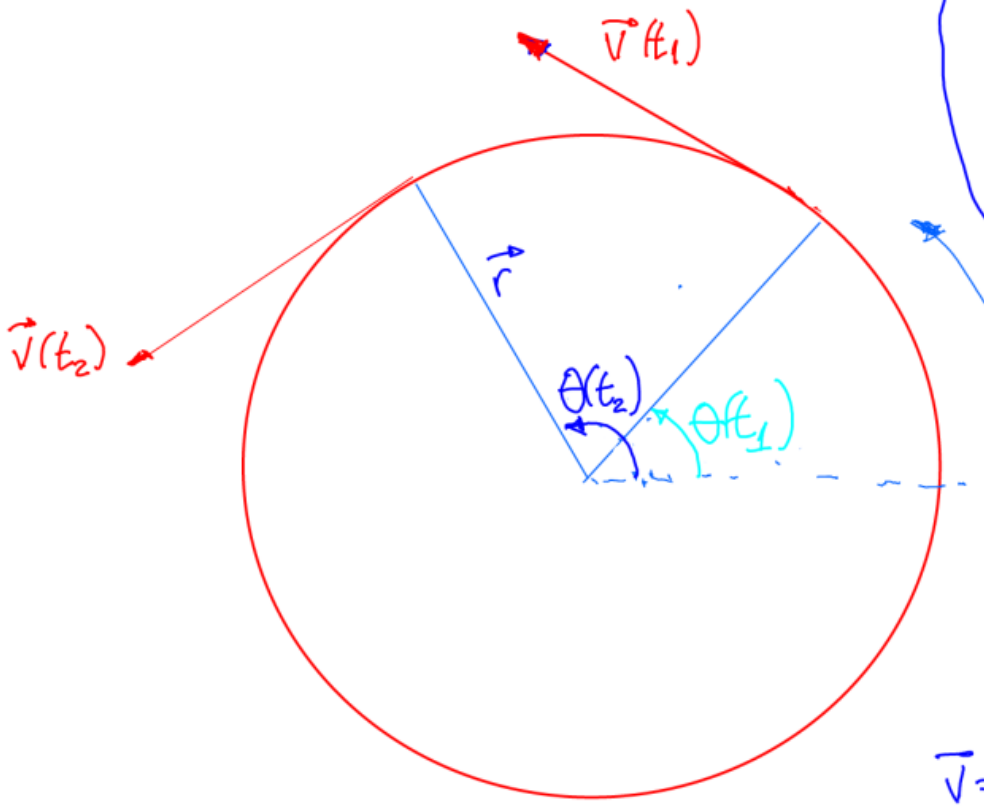
$$\hat{r} = |\hat{r}|_x\,\hat{i} + |\hat{r}|_y\,\hat{j}$$

$$\sin\theta = \frac{|\hat{\theta}|_x}{|\hat{\theta}|} = |\hat{\theta}|_x$$

$$\boxed{\hat{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}}$$

$$\cos\theta = \frac{|\hat{\theta}|_y}{|\hat{\theta}|} = |\hat{\theta}|_y$$

$$\hat{\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$



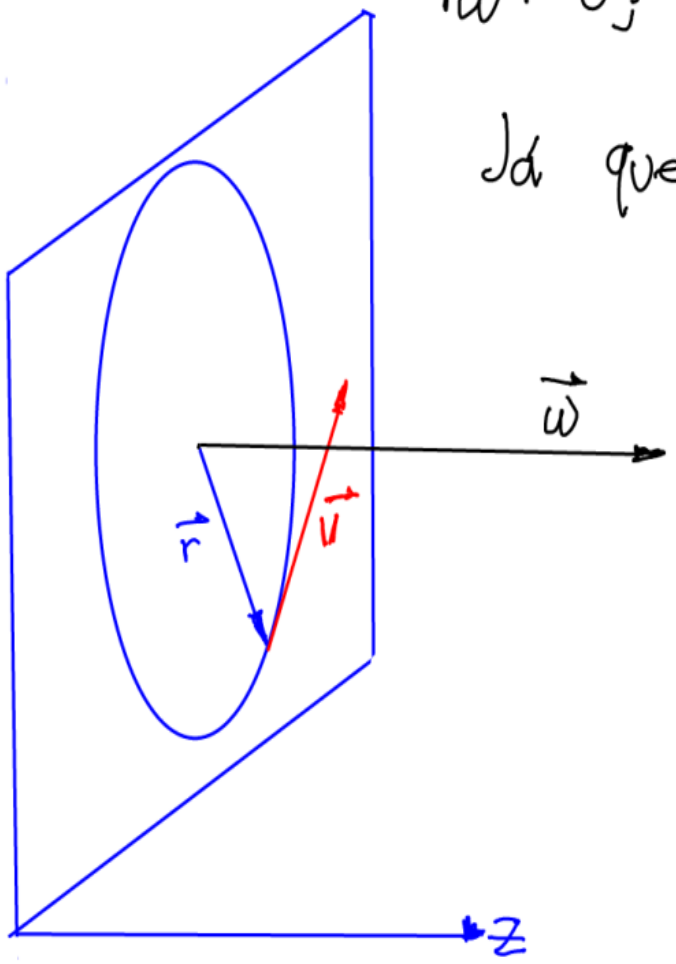
$$\hat{\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

$$\hat{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

$$\frac{d\hat{r}}{dt} = -\sin\theta\,\hat{i}\left(\frac{d\theta}{dt}\right) + \cos\theta\left(\frac{d\theta}{dt}\right)\hat{j}$$

$$\frac{d\hat{r}}{dt} = \frac{d\theta}{dt}\underbrace{\left(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j}\right)}_{\hat{\theta}}$$

$$\vec{v} = r\frac{d\hat{r}}{dt} = r\dot{\theta}\,\hat{\theta}$$



$|\vec{\omega}| = \dot{\theta}$; mas $\vec{\omega}$ como vetor é dado por $\vec{\omega} = \dot{\theta}\hat{k}$

Já que $\vec{r} \perp \vec{v} \perp \vec{\omega}$ são ortogonais entre eles

$\vec{v} = \vec{\omega} \times \vec{r}$ ou seja $\vec{v} = (\dot{\theta}\hat{k}) \times r\hat{r}$

$\vec{v} = r\dot{\theta}\,\hat{k} \times \hat{r} = r\dot{\theta}\hat{\theta}$

Em Coordenadas Polares

$\hat{r} \to 1 \qquad \hat{\theta} \to 2 \qquad \hat{k} \to 3$

$\hat{r} \times \hat{\theta} = \hat{k} \Rightarrow \underbrace{123}_{+}123$

$\hat{\theta} \times \hat{k} = \hat{r} \Rightarrow 23\underbrace{123}_{+}1$

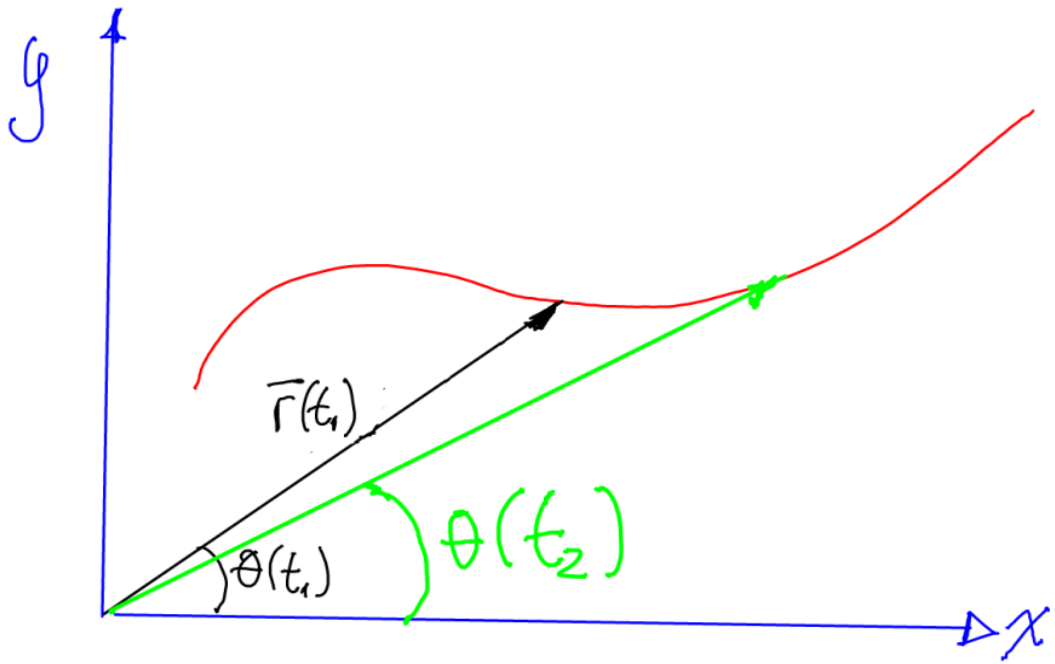
$\hat{\theta} \times \hat{r} = -\hat{k} \Rightarrow 21\underbrace{32}_{-}13$

$\hat{k} \times \hat{\theta} = -\hat{r} \Rightarrow \underbrace{321}_{-}321$

$\hat{k} \times \hat{r} = \hat{\theta} \Rightarrow 3\underbrace{123}_{+}12$

$\hat{r} \times \hat{k} = -\hat{\theta} \Rightarrow 1\underbrace{32}_{-}132$

Quando r não é fixo, é dizer no caso no qual $r = r(\theta(t))$



A velocidade é dada por $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(r\hat{r})}{dt} = \dot{r}\hat{r} + r\frac{d\hat{r}}{dt}$; mas lembremos que $\frac{d\hat{r}}{dt} = \dot{\theta}\hat{\theta}$

$$\boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}}$$

Ex: A posição de uma partícula é dada pela equação $r=3\theta$, tal que a variação do angulo em função do tempo é $\theta = 2t^3$, onde t é medido em segundos, r em metros e θ em radianes. Determine a velocidade da partícula em coordenadas polares para $\theta = 0{,}2$ rad

$r = 3(2t^3) = 6t^3$; $\theta = 2t^3$

$\dot{r} = 18t^2$ $\quad$ $\dot{\theta} = 6t^2$

Já que $\theta = 2t^3$ em $\theta = 0{,}2$ rad $\quad t = \left(\frac{0{,}2}{2}\right)^{1/3} = (0{,}1)^{1/3} = 0.464$

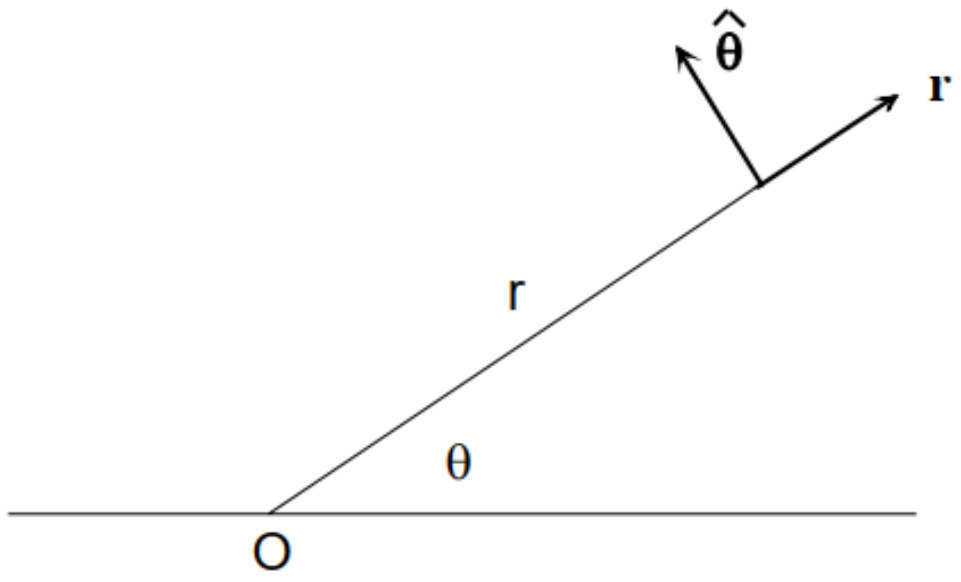
$\dot{r} = 18t^2 = 3{,}87$

$r\dot{\theta} = (6t^3)(6t^2) = 0{,}77$

$\vec{v} = 3{,}875\hat{r} + 0{,}774\hat{\theta}$

Resumo de Coordenadas polares

$$\begin{aligned}\hat{r} &= \hat{\imath}\cos\theta + \hat{\jmath}\sin\theta,\\ \hat{\theta} &= -\hat{\imath}\sin\theta + \hat{\jmath}\cos\theta.\end{aligned}$$



$$\begin{aligned}\frac{d\hat{r}}{dt} &= \dot{\theta}\left(-\hat{\imath}\sin\theta + \hat{\jmath}\cos\theta\right) = \dot{\theta}\hat{\theta},\\ \frac{d\hat{\theta}}{dt} &= -\dot{\theta}\left(\hat{\imath}\cos\theta + \hat{\jmath}\sin\theta\right) = -\dot{\theta}\hat{r}.\end{aligned}$$

velocidade angular

$$\frac{d\hat{r}}{dt} = \dot{\theta}\hat{k} \times \hat{r}$$